

Expanding linear regression models

Goals

So far we have looked at the simplest case of a regression model, in which we have one continuous X variable and one continuous Y variable. Now we want to build on that base to include

- ▶ categorical predictor variables
- ▶ models with more variables

Categorical predictors

Categorical predictors

If we have a predictor variable that is categorical rather than continuous we can still use a linear model to describe the relationship between x and y .

$$y = \beta_0 + \beta_1 * x$$

Consider a model where x is exposure to treatment and is coded 0= no treatment 1= treatment and y equals systolic blood pressure.

$$\text{bloodpressure} = \beta_0 + \beta_1 * \text{treatment}$$

then the average expected value for blood pressure among those who had treatment would be $\beta_0 + \beta_1$ and the expected value for blood pressure among those who did not have treatment would be β_0 (if treatment=0 the β_1 would be multiplied by 0)

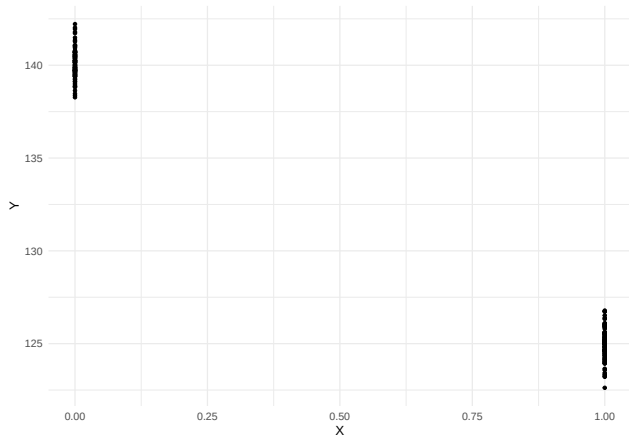
Categorical predictors

Categorical predictors

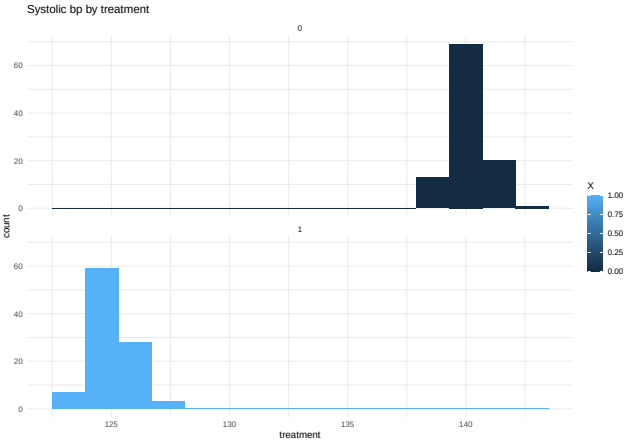
```
set.seed(241)
# here I am generating a set of x that has 200 observations and has a prob
X <- (rbinom(200, size=1, prob=.5))
# generating a normally distributed error
e<-rnorm(200, mean=0, sd=.8)
## and creating Y
Y<-140 - X*15+e

simbp <-data.frame(cbind(X,Y))
ex1<-ggplot(simbp, aes(x=X, y=Y))+
  geom_point()+
  theme_minimal(base_size=15)
```

graphically



graphically



linear model

```
lm_bp<-lm(Y ~ X, data=simbp)
tidy(lm_bp)
```

```
## # A tibble: 2 x 5
##   term          estimate std.error statistic    p.value
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)    140.      0.0777   1803. 0
## 2 X             -15.1     0.112   -135. 5.05e-197
```

```
glance(lm_bp)
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic    p.value    df logLik   AIC
##   <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl>
## 1    0.989      0.989 0.789   18290. 5.05e-197     1  -235.  477.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

linear model vs t test

```
t.test(Y ~ X, data=simbp)
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: Y by X
```

```
## t = 135.11, df = 196.37, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means between group 0 and group 1
```

```
## 95 percent confidence interval:
```

```
## 14.87010 15.31062
```

```
## sample estimates:
```

```
## mean in group 0 mean in group 1
```

```
## 140.0908 125.0005
```

binary 0/1 vs 1/2 coding

```
simbp2<-simbp%>%mutate(X2=X+1)
lm_bp2<-lm(Y ~ X2, data=simbp2)
coefficients(lm_bp)
```

```
## (Intercept)          X
##   140.09081    -15.09036
```

```
coefficients(lm_bp2)
```

```
## (Intercept)          X2
##   155.18117    -15.09036
```

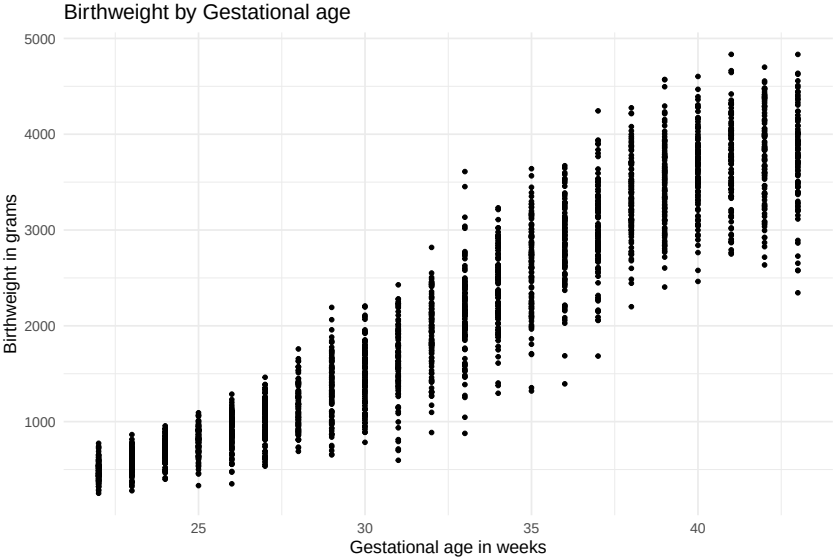
How does this change the equation? our interpretation? Why do we generally use 0/1?

Creating a binary predictor from a continuous predictor

Creating a binary predictor from a continuous predictor

Let's imagine that we are studying birthweights by gestational age. Both are collected continuously, so we might graph these data as a scatterplot. Here I will simulate a dataset based on the distribution of Canadian Female Singleton births (reference Michael S. Kramer et al. A New and Improved Population-based Canadian Reference for Birth Weight for Gestational Age.)[<http://www.pediatrics.org/cgi/content/full/108/2/e35>] and graph the results:

Graph



Linear Model for birthweight by GA

```
bwt_ga_cont<-lm(bwt~ GA, data=brths)
tidy(bwt_ga_cont)
```

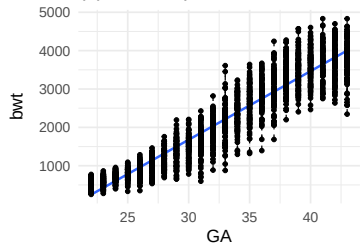
```
## # A tibble: 2 x 5
##   term          estimate std.error statistic p.value
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)   -3689.    43.7     -84.4      0
## 2 GA             179.     1.32     135.      0
```

```
glance(bwt_ga_cont)
```

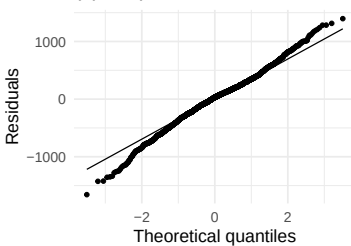
```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic p.value    df logLik   AIC
##   <dbl>      <dbl> <dbl>    <dbl>    <dbl> <dbl> <dbl> <dbl>
## 1    0.893    0.893  393.    18351.      0      1 -16262. 32531.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Model fit

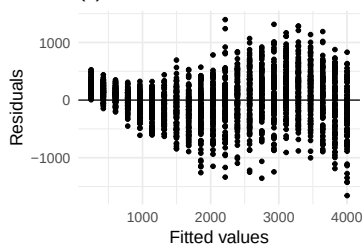
(a) Scatter plot



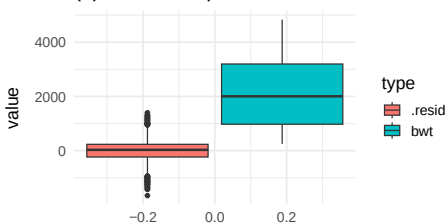
(b) QQplot



(c) Fitted vs. residuals

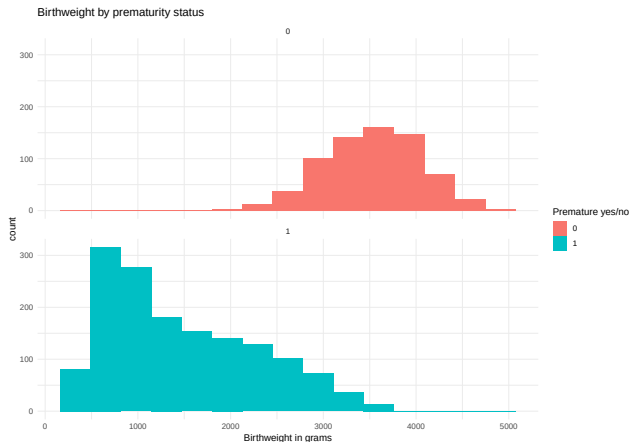


(d) Amount explained



What happens if we categorize?

We often look at birth outcomes by premature status. If I create a 0/1 indicator variable for gestational age less than 37 weeks the distribution looks like this:



linear model

```
## # A tibble: 2 x 5
```

##	term	estimate	std.error	statistic	p.value
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	(Intercept)	3526.	27.4	129.	0
## 2	premature	-2055.	33.2	-61.9	0

```
## # A tibble: 1 x 12
```

##	r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC
##	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	0.636	0.636	725.	3835.	0	1	-17611.	35227.

```
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

t-test

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: bwt by premature
```

```
## t = 71.803, df = 1970, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means between group 0 and group 1
```

```
## 95 percent confidence interval:
```

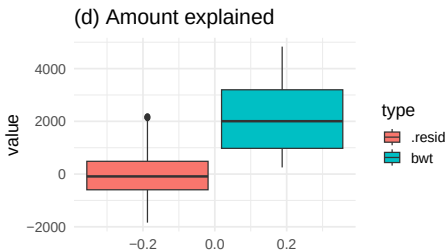
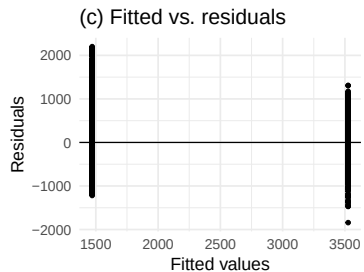
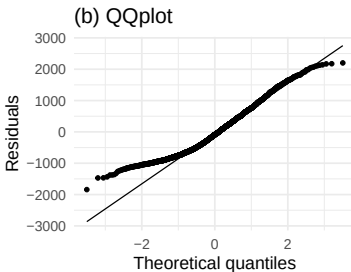
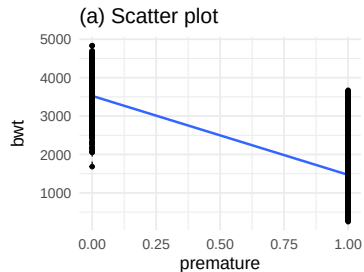
```
## 1999.312 2111.594
```

```
## sample estimates:
```

```
## mean in group 0 mean in group 1
```

```
## 3525.912 1470.459
```

Model fit for binary predictor



Categorical predictors with more than 2 levels

Adding categories

Remember that a coefficient in a linear model represents the change in Y related to a one unit change in X , so how do we handle this for categories? Even in ordinal variables - those that have an inherent order - we generally do not expect the difference between each category to be the same. For nominal variable we would be even less comfortable with that structure. . . .so

Adding categories

As long as the categories are mutually exclusive and exhaustive we can create a series of “dummy” variables coded 0/1 that represent the categories.

$$y = \beta_0 + \beta_1 * category_1 + \beta_2 * category_2 \cdots + \beta_j * category_{k-1}$$

where k is the number of categories. Why do we have 1 fewer terms than categories? Let's look at an example. . .

Example: Butterfat

This dataset contains data on the average butterfat content of milk from random samples of dairy cattle. Data came from Canadian records of pure-bred dairy cattle

```
data(butterfat, package="faraway")  
str(butterfat)
```

```
## 'data.frame':    100 obs. of  3 variables:  
## $ Butterfat: num  3.74 4.01 3.77 3.78 4.1 4.06 4.27 3.94 4.11 4.25 ...  
## $ Breed : Factor w/ 5 levels "Ayrshire","Canadian",...: 1 1 1 1 1 1 ...  
## $ Age : Factor w/ 2 levels "2year","Mature": 2 1 2 1 2 1 2 1 2 1
```


Example: Butterfat

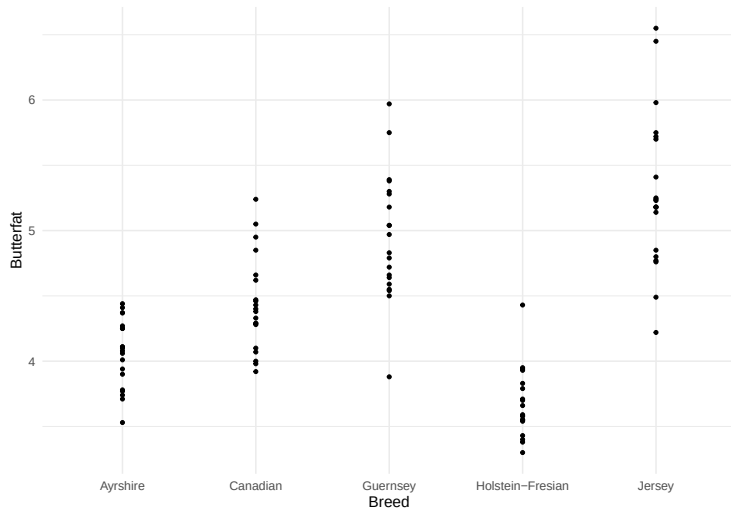
We can look at the means by group here

```
butterfat%>%  
  group_by(Breed) %>%  
  summarize(mean_fat = mean(Butterfat, na.rm=TRUE))
```

```
## # A tibble: 5 x 2  
##   Breed          mean_fat  
##   <fct>         <dbl>  
## 1 Ayrshire      4.06  
## 2 Canadian     4.44  
## 3 Guernsey     4.95  
## 4 Holstein-Fresian 3.67  
## 5 Jersey       5.29
```

Example: Butterfat

and graph them here:



Example: Butterfat

What does the model output look like?

```
lm_fat<-lm(Butterfat ~ Breed, data=butterfat)
tidy(lm_fat)
```

```
## # A tibble: 5 x 5
```

##	term	estimate	std.error	statistic	p.value
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	(Intercept)	4.06	0.0928	43.7	9.51e-65
## 2	BreedCanadian	0.378	0.131	2.88	4.86e- 3
## 3	BreedGuernsey	0.89	0.131	6.78	1.01e- 9
## 4	BreedHolstein-Fresian	-0.390	0.131	-2.98	3.71e- 3
## 5	BreedJersey	1.23	0.131	9.39	3.33e-15

where is the expected value for Ayrshire?

choosing a different reference category

```
butterfat_relevel <- within(butterfat, Breed <- relevel(Breed, ref = "Holstein"))  
  
lm_fat2 <- lm(Butterfat ~ Breed, data=butterfat_relevel)  
  
coefficients(lm_fat)
```

##	(Intercept)	BreedCanadian	BreedGuernsey
##	4.0600	0.3785	0.8900
##	BreedHolstein-Friesian	BreedJersey	
##	-0.3905	1.2325	

```
coefficients(lm_fat2)
```

##	(Intercept)	BreedAyrshire	BreedCanadian	BreedGuernsey	BreedJersey
##	3.6695	0.3905	0.7690	1.2805	1.6230

models with more than one predictor variable

Regression line with multiple x variables

In the simple case our regression line was:

$$y = \beta_0 + \beta_1 * x$$

in the case of multiple variables we can write this more generally as:

$$y = \beta_0 + \beta_1 * x_1 + \beta_2 * x_2 + \cdots + \beta_p * x_p$$

where p is the number of x variables included in the model.

interpreting the coefficients

In a simple model, we interpreted the β or slope coefficient as the change in the predicted or “expected” value of y associated with a one unit change in x .

When there are multiple x variables, the interpretation is that the coefficient represents the change in the expected value of y associated with a one unit change in x , **holding all other factors in the model constant** or “controlling for” the other factors in the model.

Example: Butterfat

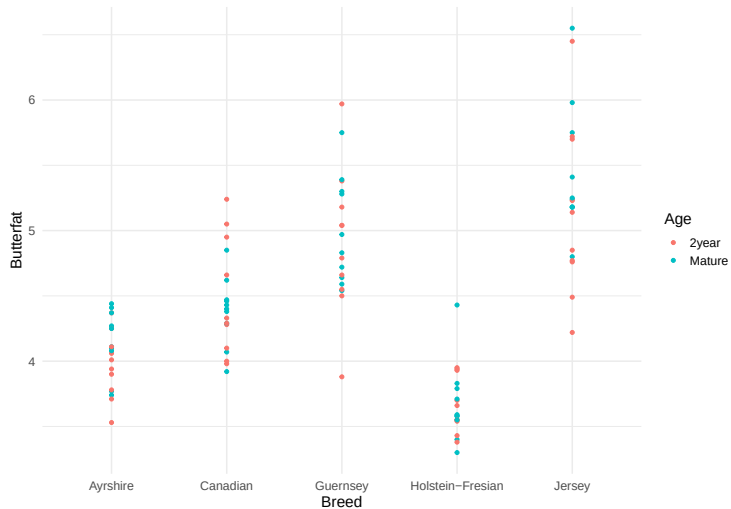
Let's say we also suspect that there is a difference in butterfat content as cows reach maturity

```
butterfat%>%  
  group_by(Age) %>%  
  summarize(mean_fat = mean(Butterfat, na.rm=TRUE))
```

```
## # A tibble: 2 x 2  
##   Age      mean_fat  
##   <fct>      <dbl>  
## 1 2year      4.43  
## 2 Mature    4.53
```


Example: Butterfat

Including age in the visualization by breed



Butterfat model with both factors

```
lm_fat3<-lm(Butterfat ~ Breed + Age, data=butterfat_relevel)
tidy(lm_fat3)
```

```
## # A tibble: 6 x 5
```

##	term	estimate	std.error	statistic	p.value
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	(Intercept)	3.62	0.101	35.7	1.88e-56
## 2	BreedAyrshire	0.390	0.131	2.98	3.62e- 3
## 3	BreedCanadian	0.769	0.131	5.88	6.31e- 8
## 4	BreedGuernsey	1.28	0.131	9.79	5.23e-16
## 5	BreedJersey	1.62	0.131	12.4	1.68e-21
## 6	AgeMature	0.105	0.0828	1.26	2.09e- 1

Butterfat model with both factors

$$\textit{Butterfat} = \beta_0 + \beta_1 * \textit{Breed}_1 + \beta_2 * \textit{Breed}_2 + \beta_3 * \textit{Breed}_3 + \beta_4 * \textit{Breed}_4 + \beta_5 * \textit{Matureage}$$

$$= 3.62 + 0.39 * \textit{Ayrshire} + 0.7690 * \textit{Canadian} + 1.28 * \textit{Guernsey} + 1.62 * \textit{Jersey} + 0.10 * \textit{Matureage}$$

What is β_0 in this model?

comparing any 2 groups

What if we are interested in the difference in butterfat between milk from a young Ayrshire cow and a mature Guernsey?

Young Ayrshire:

$$E[\text{Butterfat} | \text{Breed} = \text{Ayrshire}, \text{Age} = \text{young}] = 3.62 + 0.39 * \text{Ayrshire} = 4.01$$

Mature Guernsey:

$$E[\text{Butterfat} | \text{Breed} = \text{Guernsey}, \text{Age} = \text{Mature}] = 3.62 + 1.28 * \text{Guernsey} + 0.10 * \text{Mature} = 5$$

$$\begin{aligned} E[\text{Butterfat} | \text{Breed} = \text{Ayrshire}, \text{Age} = \text{young}] - E[\text{Butterfat} | \text{Breed} = \text{Guernsey}, \text{Age} = \text{Mature}] \\ = 4.01 - 5 = -0.99 \end{aligned}$$

So the difference is 0.99 less butterfat among young Ayrshire cows compared to mature Guernseys

testing for variables with multiple categories

When we are looking at variables that are continuous or binary [0,1] the t-test values for each β give us a sense of the importance of each variable.

Why is this a problem in a variable with multiple categories?

```
## # A tibble: 6 x 5
##   term                estimate std.error statistic  p.value
##   <chr>              <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)        3.62      0.101     35.7 1.88e-56
## 2 BreedAyrshire      0.390     0.131      2.98 3.62e- 3
## 3 BreedCanadian      0.769     0.131      5.88 6.31e- 8
## 4 BreedGuernsey       1.28     0.131      9.79 5.23e-16
## 5 BreedJersey         1.62     0.131     12.4 1.68e-21
## 6 AgeMature           0.105     0.0828     1.26 2.09e- 1
```

comparing models f-test

We will use the `anova()` function in `r` to compare a model that includes Breed to one that does not

```
lm_fat4<-lm(Butterfat ~ Age, data=butterfat_relevel)
```

```
anova(lm_fat4, lm_fat3)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Model 1: Butterfat ~ Age
```

```
## Model 2: Butterfat ~ Breed + Age
```

```
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
```

```
## 1      98 50.415
```

```
## 2      94 16.094  4    34.321 50.115 < 2.2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

comparing models R-squared

We might also be interested in comparing the amount of variability explained by our two models:

```
glance(lm_fat4)
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic p.value    df logLik   AIC    BIC
##   <dbl>      <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1  0.00540    -0.00475 0.717     0.532   0.468     1 -108.  221.  222.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
glance(lm_fat3)
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic p.value    df logLik   AIC    BIC
##   <dbl>      <dbl> <dbl>     <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1  0.682      0.666 0.414     40.4 5.51e-22     5 -50.6  115.  116.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

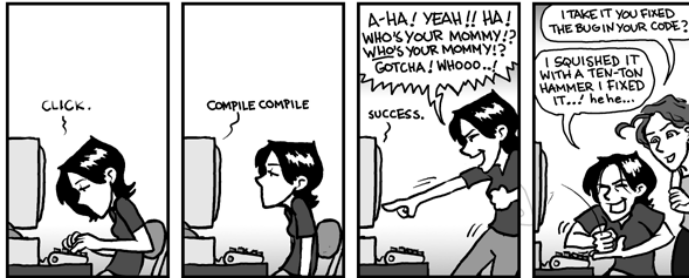
Parting humor

1/14/22, 8:00 AM

Piled Higher and Deeper

Piled Higher and Deeper by *Jorge Cham*

www.phdcomics.com



phd.stanford.edu/

title: "Squished it with a ten-ton hammer I fixed it. Bugs. Code." - originally published 4/28/2000